



Lecture 16

- Laplace Transform Circuit Analysis



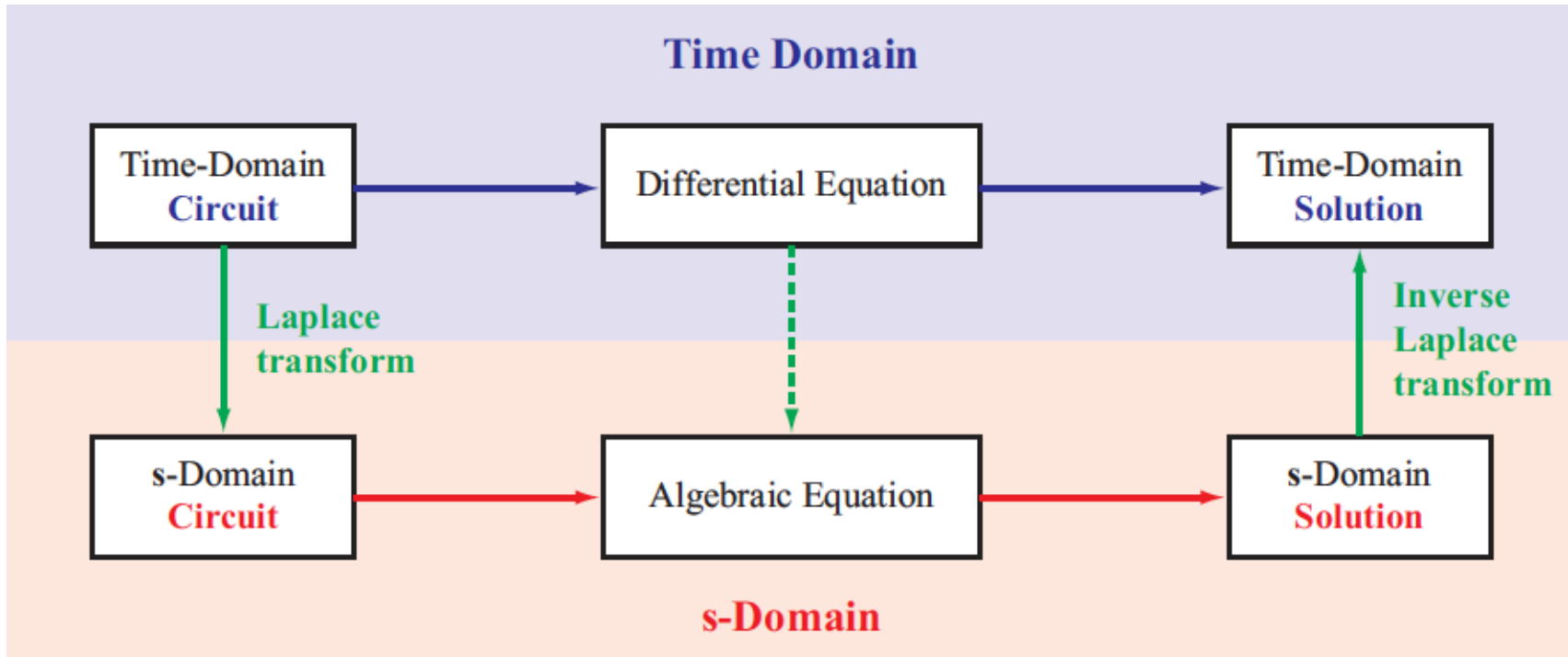
Analysis Techniques

Circuit Excitation	Method of Solution
dc	DC/Transient analysis
ac	Phasor-domain analysis (Steady state only)
Periodic waveform	Fourier series + Phasor-domain (Steady state only)
Waveform (single-sided)	Laplace transform (transient + steady state)

Single-sided: defined over $[0, \infty]$



Laplace Transform Technique





Application to Differential Equations

- The Laplace transform is useful in solving linear integrodifferential equations.
 - Initial conditions are automatically taken into account.

Use the Laplace transform to solve the differential equation

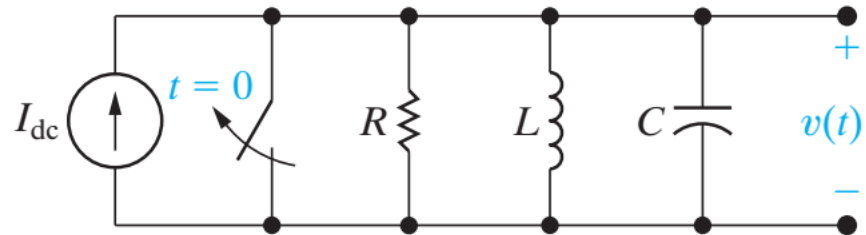
$$\frac{d^2v(t)}{dt^2} + 6\frac{dv(t)}{dt} + 8v(t) = 2u(t)$$

subject to $v(0) = 1, v'(0) = -2$.

$$[s^2V(s) - sv(0) - v'(0)] + 6[sV(s) - v(0)] + 8V(s) = \frac{2}{s}$$

$$V(s) = \frac{s^2 + 4s + 2}{s(s+2)(s+4)} = \frac{\frac{1}{4}}{s} + \frac{\frac{1}{2}}{s+2} + \frac{\frac{1}{4}}{s+4} \quad v(t) = \frac{1}{4}(1 + 2e^{-2t} + e^{-4t})u(t)$$

Applying the Laplace Transform



$$\frac{v(t)}{R} + \frac{1}{L} \int_0^t v(x) dx + C \frac{dv(t)}{dt} = I_{dc} u(t).$$

$$\frac{V(s)}{R} + \frac{1}{L} \frac{V(s)}{s} + C[sV(s) - v(0^-)] = I_{dc} \left(\frac{1}{s} \right)$$

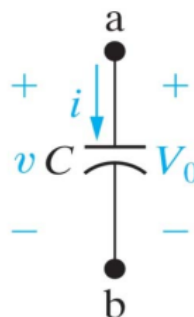


V-I relations of R

$$U_R(s) = RI_R(s)$$

S-domain circuit models for a capacitor

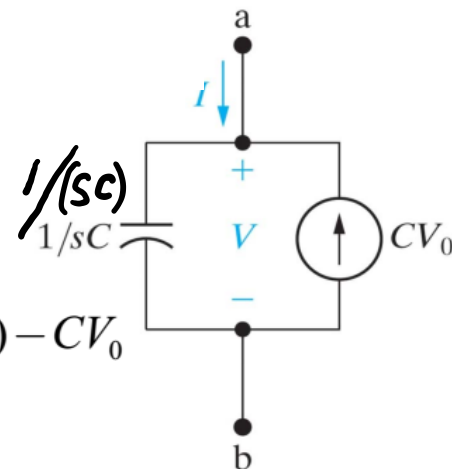
$$I_C(s) = sCU_C(s) - Cu_C(0_-)$$



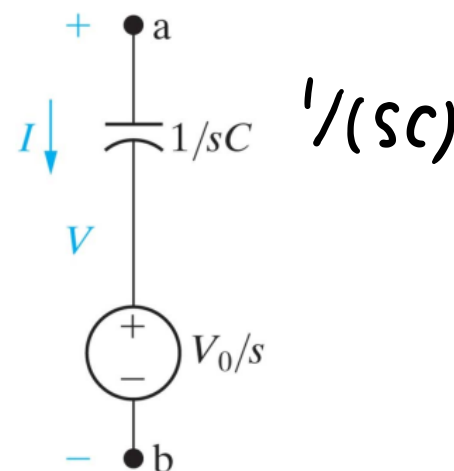
$$i(t) = C \frac{dv(t)}{dt}$$



$$I(s) = sCV(s) - CV_0$$

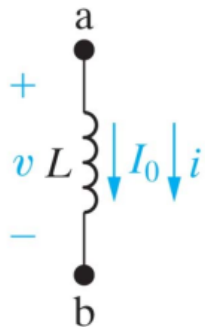


$$V(s) = \frac{1}{sC} I(s) + \frac{V_0}{s}$$



S-domain circuit models for an inductor

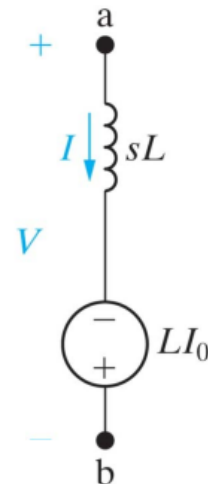
$$I_L(s) = \frac{i_L(0_-)}{s} + \frac{1}{sL} U_L(s)$$



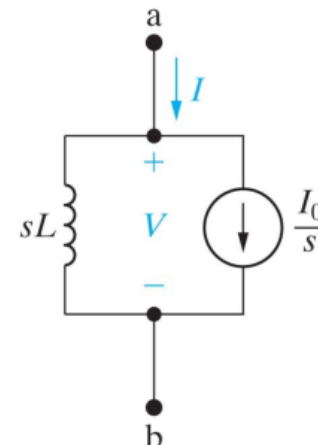
$$v(t) = L \frac{di(t)}{dt}$$



$$V(s) = sLI(s) - LI_0$$



$$I(s) = \frac{1}{sL} V(s) + \frac{I_0}{s}$$

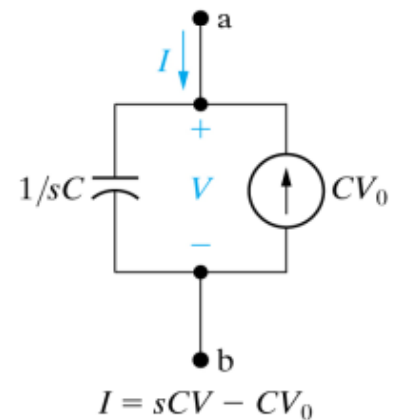
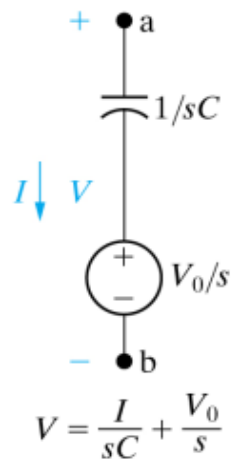
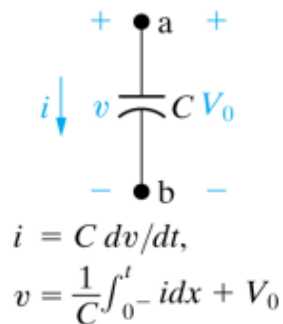
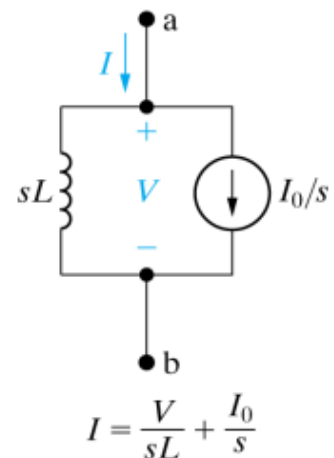
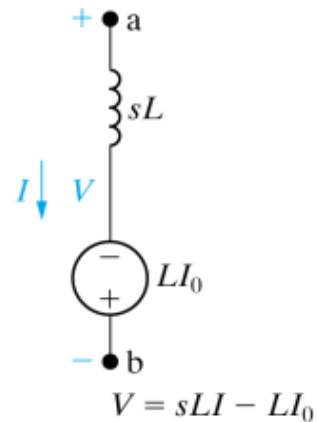
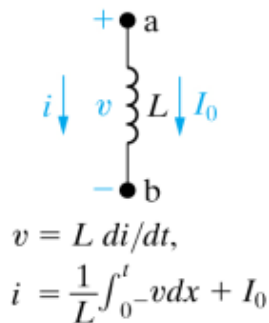
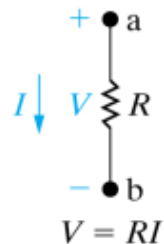
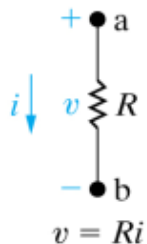




Summary

Time domain

s-domain





Dependent Sources

- The models for dependent sources are easy to develop, drawing from the simple fact that if the Laplace transform of $f(t)$ is $F(s)$, then the Laplace transform of $af(t)$ is $aF(s)$ — the linearity property.

$$\mathcal{L}[av(t)] = aV(s)$$

$$\mathcal{L}[ai(t)] = aI(s)$$

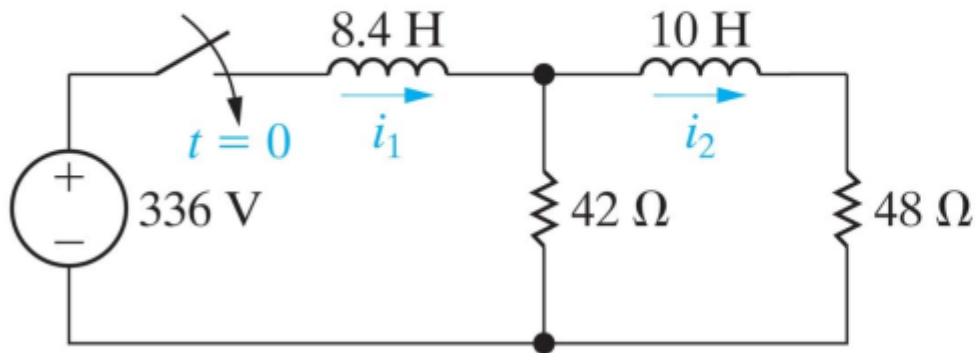


Steps in Applying the Laplace transform

- Transform the circuit from the time domain to the Laplace (s) domain, including possible initial conditions.
- Solve the circuit using nodal analysis, mesh analysis, source transformation, superposition, or any other analysis technique with which we are familiar.
- Take the inverse transform of the solution and thus obtain the solution in the time domain.

Example 1

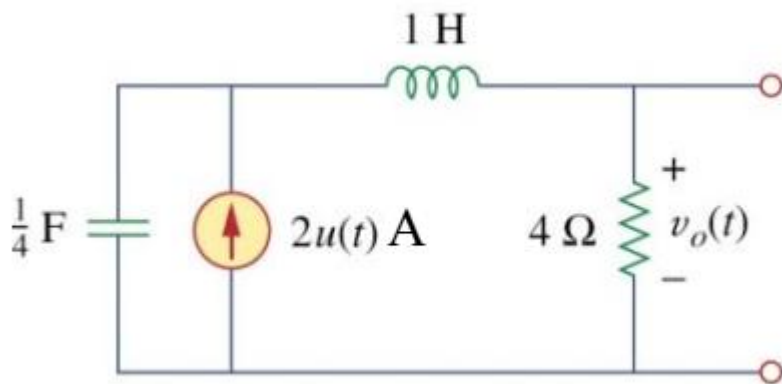
Assuming no initial energy storage, find $i_1(t)$ and $i_2(t)$ for $t > 0$.





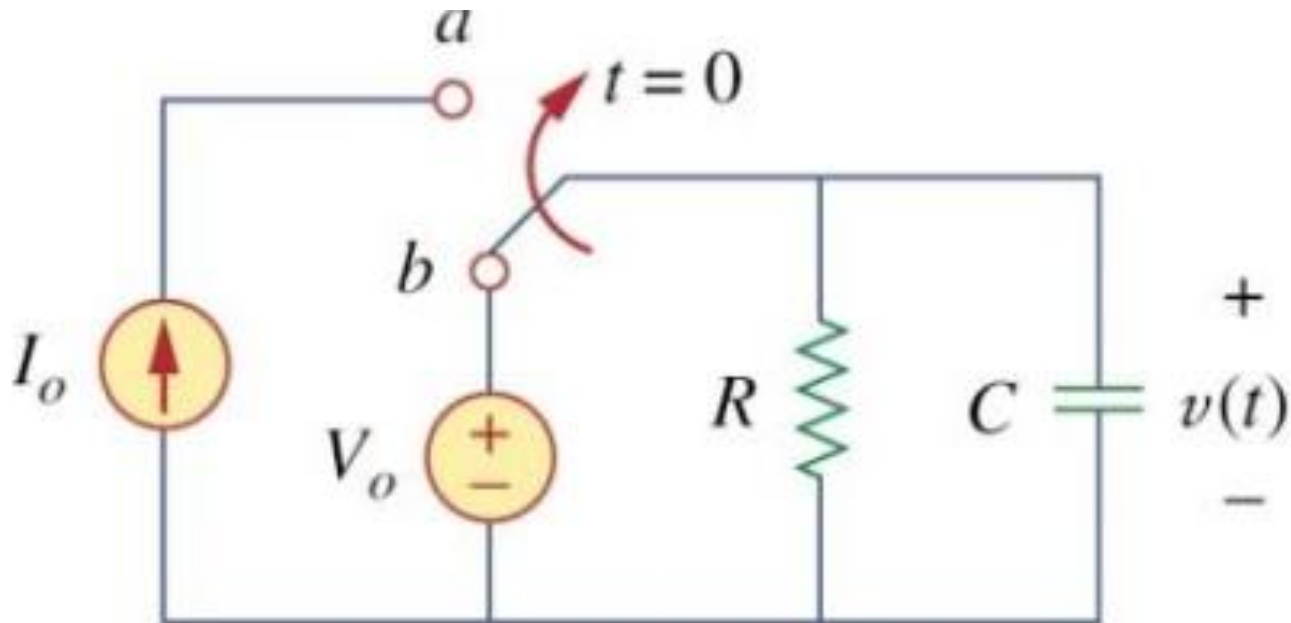
Example 2

Determine $v_o(t)$ for $t > 0$ assuming zero initial conditions:



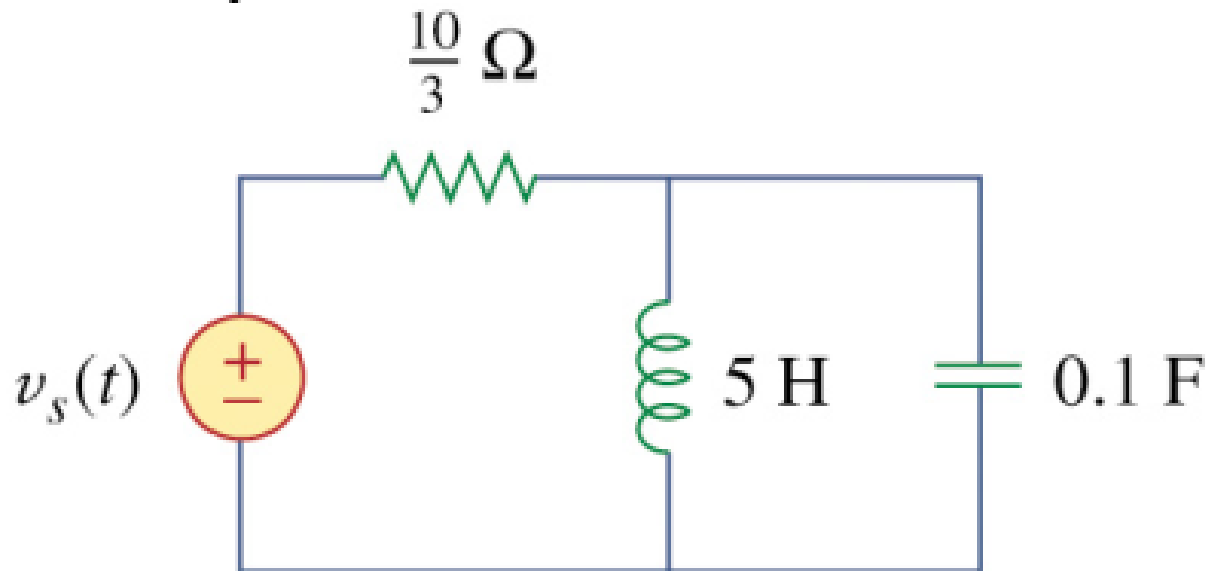
Example 3

- The switch has been in position b for a long time. It is moved to position a at $t = 0$. Determine $v(t)$ for $t > 0$.



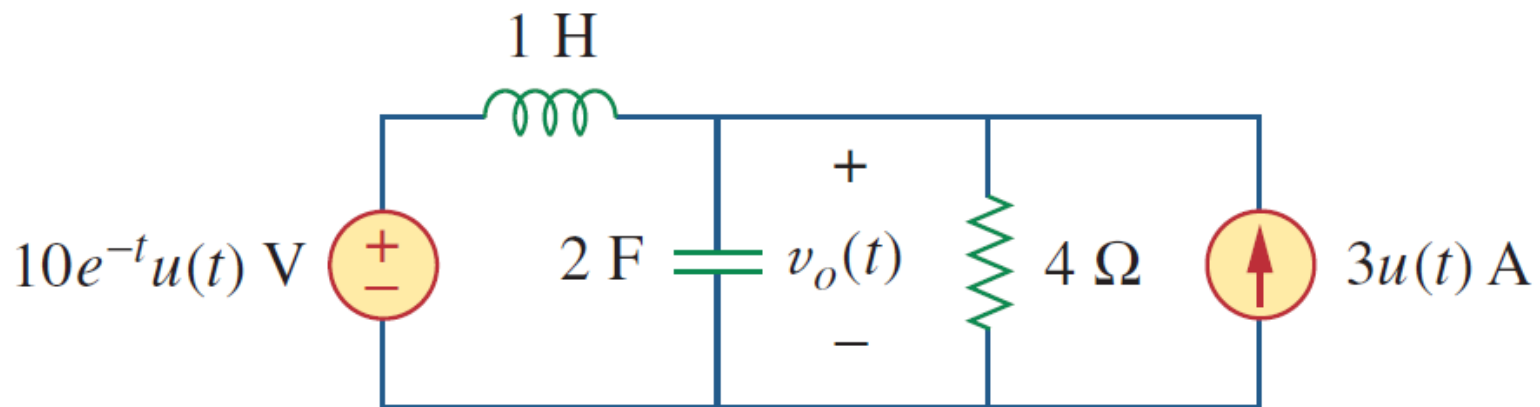
Example 4

- Find the voltage across the capacitor assuming that $v_s(t) = 10u(t)$ V, and assume that at $t = 0$, -1 A flows through the inductor and $+5$ V is across the capacitor.



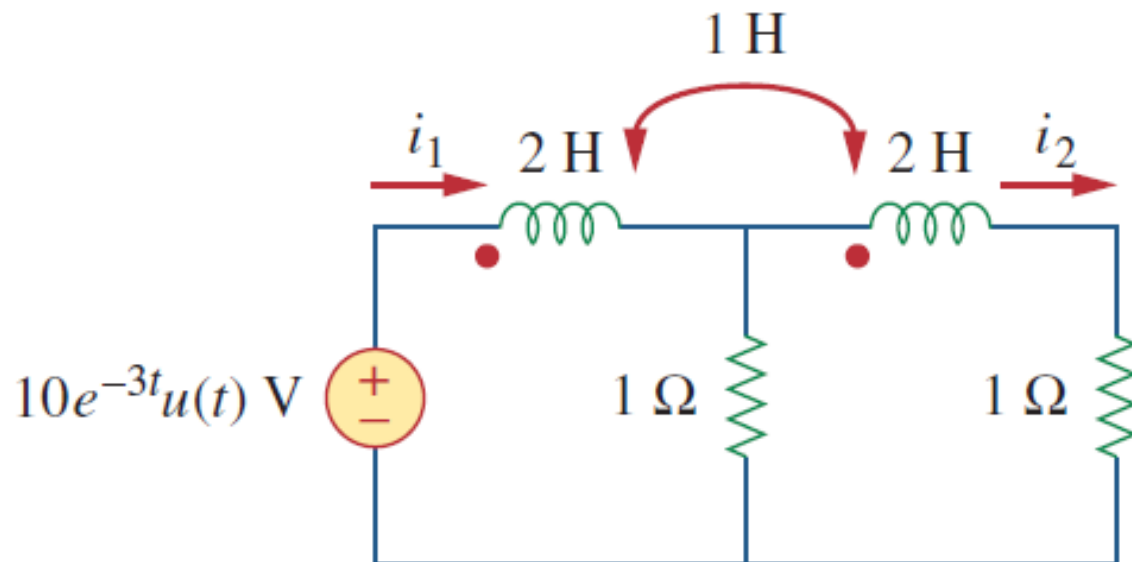
Example 5

16.35 Find $v_o(t)$ in the circuit of Fig. 16.58.



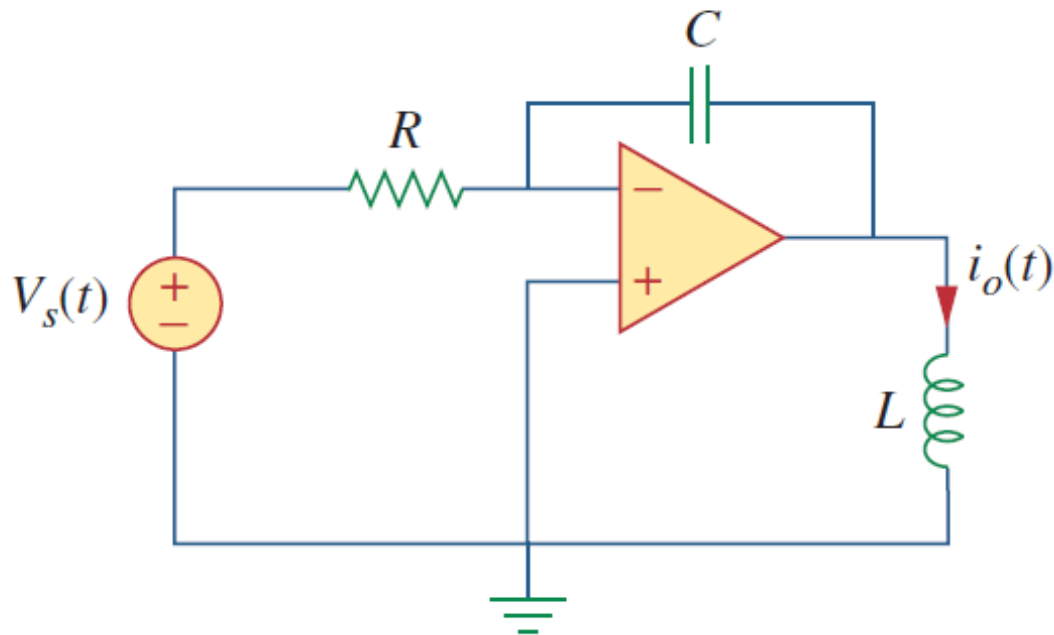
Example 6

16.69 Find $I_1(s)$ and $I_2(s)$ in the circuit of Fig. 16.92.



Example 7

16.81 For the op-amp circuit in Fig. 16.99, find the transfer function, $T(s) = I(s)/V_s(s)$. Assume all initial conditions are zero.



- The switch in the circuit below has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$ using the Laplace transform.

